Guaranteed Location of More Source than Sensors ... QH

$R = APA^H + \sigma^2 I$

$r = (A^* \otimes A)\, vec(P) + \sigma^2 vec(I)$

$= (A^* \odot A)\, p + \sigma^2 vec(I)$

$\underset{M^2 \times N}{A_{KR}} = A^* \odot A \qquad [A_{KR}]_{(m_1, m_2), n} = e^{j2\pi \underbrace{(d_{m_1} - d_{m_2})}_{\in \mathbb{D}} n/N}$

2nd order, BP

$\min \|p\|_0 \quad s.t.\ r = A_{KR} p$ 要求 $2k \le rank(A_{KR})$

$\mathbb{S} = \{d_m\} \quad |\mathbb{S}| = M$

$\mathbb{D} = \{d_{m_1} - d_{m_2}\}$ virtual array / difference co-array

consecutive $O(M^2)$

ULA, $\mathbb{D} = -M+1 \sim M+1$

$rank(A_{KR}) = |\mathbb{D}| = 2M-1$ 故 $k \le M$

Nested, $\mathbb{S} = 0 \sim \frac{M}{2}-1 \ \cup\ m(\frac{M}{2}+1)-1,\ 1 \le m \le \frac{M}{2}$

$\mathbb{D} = -(\frac{M^2}{4} + \frac{M}{2} - 1) \sim \frac{M^2}{4} + \frac{M}{2} - 1$

$rank(A_{KR}) = |\mathbb{D}| = \frac{M^2}{2} + M - 1$

e.g. $M = 6$, $\mathbb{S} = \{\begin{matrix} 0.1.2 \\ 3.7.11 \end{matrix}\}$, $\mathbb{D} = -11 \sim 11$

Coprime, $\mathbb{S} = M_1 m,\ 0 \le m \le M_2 - 1 \ \cup\ M_2 m,\ 0 \le m \le 2M_1 - 1$

$|\mathbb{S}| = M = 2M_1 + M_2 - 1$

$\mathbb{D} \supseteq -M_1 M_2 \sim M_1 M_2$

e.g. $M_1 = 2$, $M_2 = 3$, $\mathbb{S} = \{\begin{matrix} 0.2.4 \\ 0.\ 3.\ 6.\ 9 \end{matrix}\}$

$\mathbb{D} = \{-9.\ \cancel{-8}.\ \underbrace{-7.-6 \sim 6.}\ 7.\ \cancel{8}.9\}$

连续部分 $\mathbb{U} = -M_{ca} \sim M_{ca}$